

Autoencoders A-Z Theory Guide

Learning Day 6 - Beginner-Friendly Theory, Examples, Sources, Visuals, and Glossary

This document explains the theory behind the three completed notebooks and includes the missing tutor-theory topics. The remaining supplementary practical notebooks will be created later, but the theory is included here so the conceptual coverage is complete.

1. What Is an Autoencoder?

An autoencoder is a neural network trained to reconstruct its own input. It first compresses the input into a smaller representation and then rebuilds the original input from that representation.

Dense Autoencoder Flow

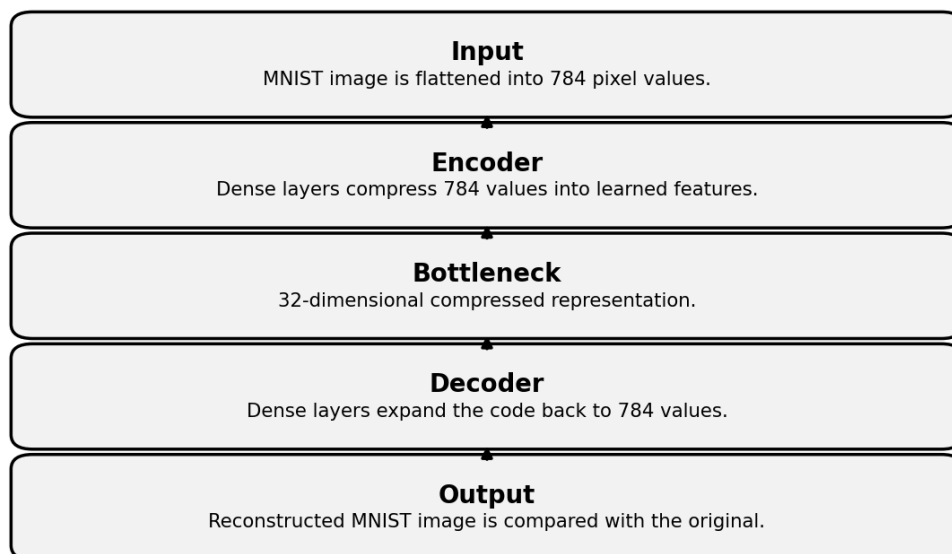


Figure 1: Basic autoencoder compression and reconstruction idea.

Simple example: a handwritten digit image starts as 784 pixel values. The encoder compresses it into fewer values, and the decoder tries to rebuild the same digit. If the rebuilt image is close to the original, reconstruction loss is low.

Source note: Tutor Autoencoder PDF/PPT describes autoencoders as unsupervised neural networks that reconstruct input data through encoding and decoding.

2. Core Parts of an Autoencoder

Part	Simple Meaning	Notebook Example
Input layer	Receives original data.	MNIST/Fashion-MNIST image.
Encoder	Compresses input into a smaller	Dense layers or Conv2D layers.

Part	Simple Meaning	Notebook Example
	representation.	
Bottleneck / latent space	Small hidden representation of the input.	32 values, 7x7x16 feature maps, or 2D VAE latent space.
Decoder	Reconstructs the input from the compressed representation.	Dense decoder, Conv decoder, or VAE decoder.
Output layer	Produces reconstructed version of the input.	784 pixels or 28x28 image.

3. Reconstruction Loss

Reconstruction loss measures how different the reconstructed output is from the original input. Lower reconstruction loss means the autoencoder rebuilt the input more accurately.

Loss	Simple Meaning	When Useful
MSE	Average squared mistake. Big errors are punished strongly.	Images, signals, numeric reconstruction.
MAE	Average absolute mistake. More direct and less aggressive than MSE.	Robust direct error measurement.
Binary Cross-Entropy	Measures reconstruction quality for 0-1 outputs.	Normalised image pixels with sigmoid output.

Difference between MSE and MAE: if the mistake is 5, MAE counts 5 but MSE counts 25. This is why MSE punishes large mistakes much more strongly.

4. Dense Autoencoder

A Dense Autoencoder uses fully connected Dense layers. It usually requires images to be flattened. In the MNIST notebook, each image became 784 values and was compressed to a 32-dimensional bottleneck.

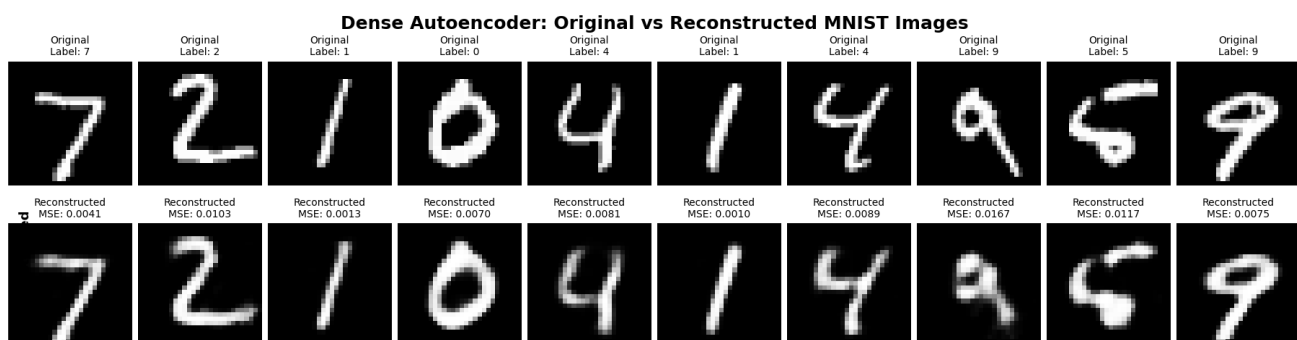


Figure 2: Dense Autoencoder reconstruction example.

Strength: easy to understand and useful for learning the basic autoencoder workflow. Weakness: flattening loses 2D image structure, so reconstructions can become blurry.

5. Convolutional Autoencoder

A Convolutional Autoencoder uses Conv2D layers, pooling, and upsampling. It keeps the image structure as 28 x 28 x 1, so it can learn local patterns such as edges, textures, soles, sleeves, and object outlines.

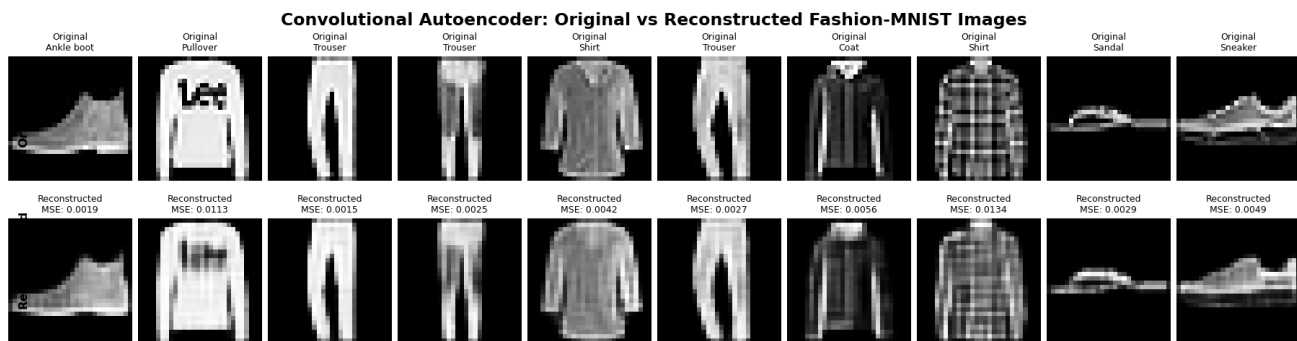


Figure 3: Conv Autoencoder reconstruction example.

The Conv Autoencoder performed much better than the Dense baseline on Fashion-MNIST, reducing average reconstruction error by about 82.3% on the comparison set.

6. Feature Extraction with Autoencoders

The encoder can be used as a feature extractor. It converts raw input, such as pixels, into compressed learned features. These features can be used for clustering, visualisation, similarity search, classification input, or anomaly detection.

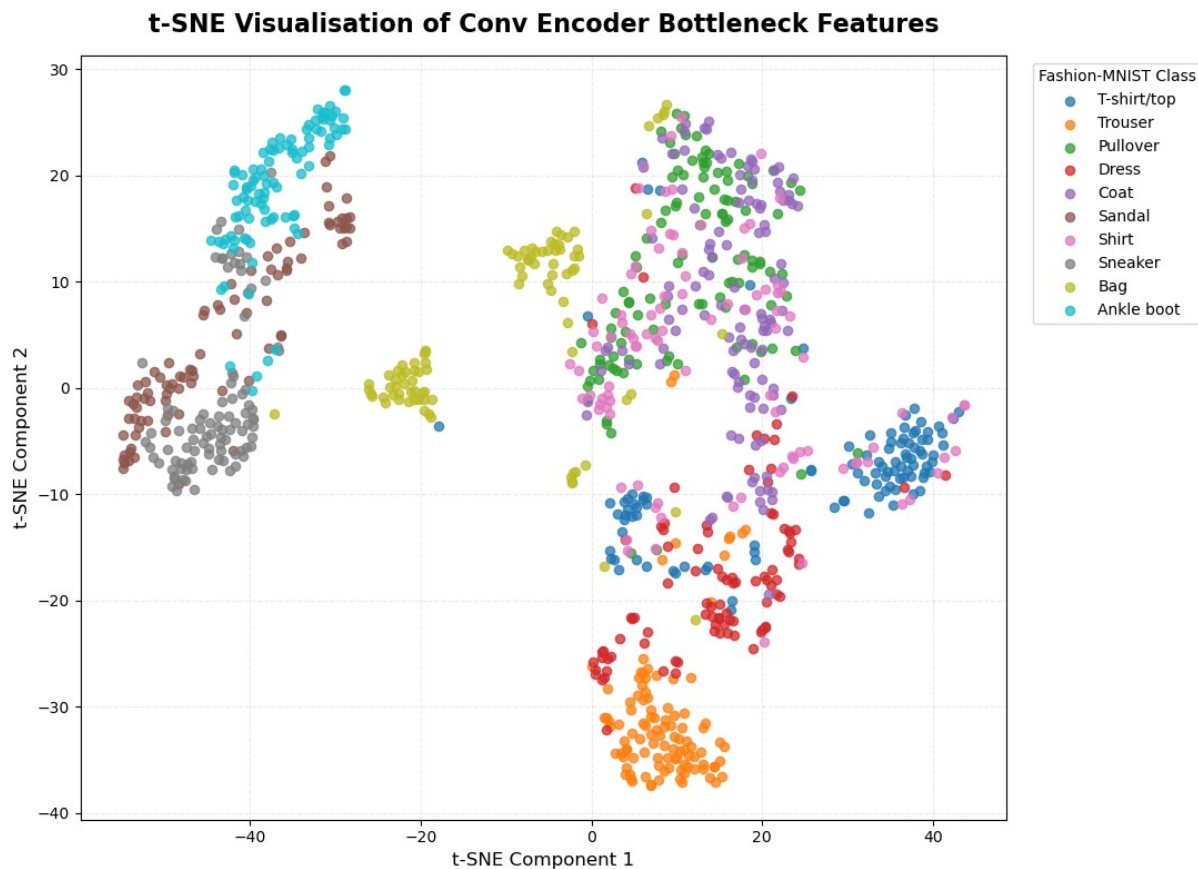


Figure 4: t-SNE plot of Conv Encoder features.

In the intermediate notebook, t-SNE and UMAP showed that some categories grouped together, especially trousers, bags, and footwear. Upper-body clothing overlapped more because those classes look visually similar.

7. Variational Autoencoder

A VAE is a generative autoencoder. Instead of encoding an image into one fixed code, it encodes the image into a probability distribution using `z_mean` and `z_log_var`. A sampled point `z` is then passed to the decoder.

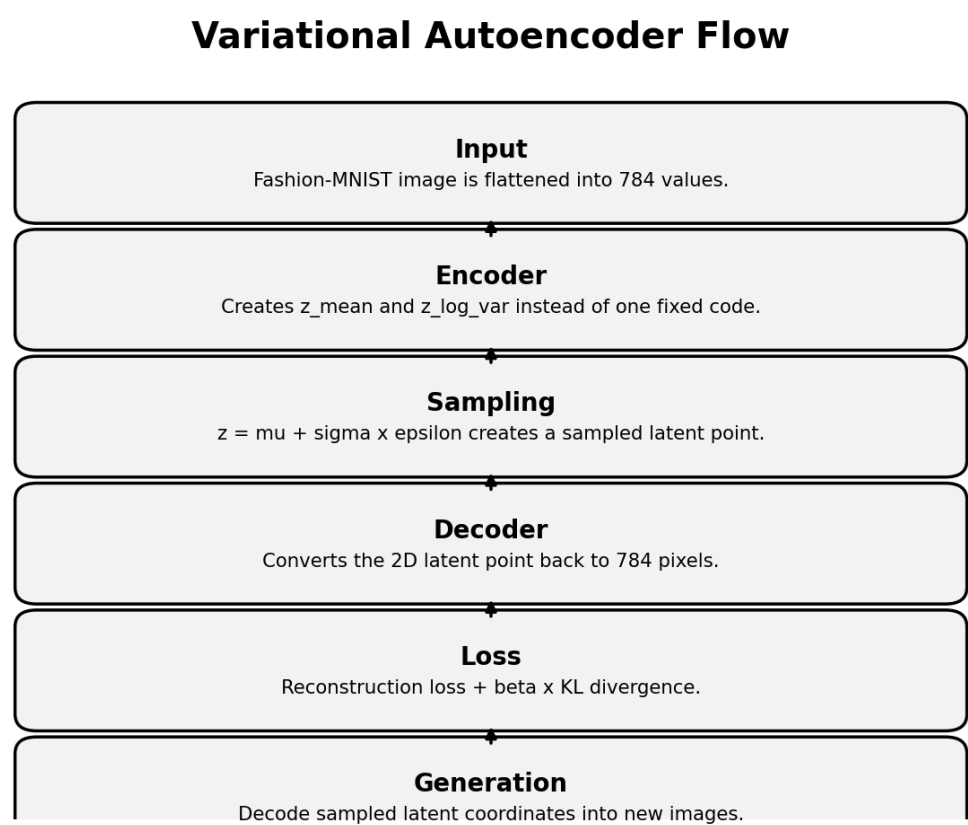


Figure 5: VAE flow with `z_mean`, `z_log_var`, `z`, decoder, and loss.

VAE Term	Simple Meaning
<code>z_mean</code>	Centre position of the image in latent space.
<code>z_log_var</code>	Spread or uncertainty around that centre.
<code>epsilon</code>	Random noise used for sampling.
<code>z</code>	Actual sampled latent point passed to the decoder.
KL divergence	Loss term that keeps latent space smooth and close to a normal distribution.

The reparameterisation trick is $z = \mu + \sigma \times \epsilon$. It lets the VAE use randomness while still allowing training through backpropagation.

8. beta-VAE

A beta-VAE changes the strength of KL divergence using the formula: $\text{total_loss} = \text{reconstruction_loss} + \beta \times \text{KL_loss}$. In the expert notebook, $\beta = 1$ and $\beta = 4$ were compared.

Model	Reconstruction MSE	KL Loss	Interpretation
$\beta = 1$	0.039027	7.490874	Better reconstruction quality.
$\beta = 4$	0.044348	3.894046	Stronger latent regularisation, blurrier

Model	Reconstruction MSE	KL Loss	Interpretation
			images.

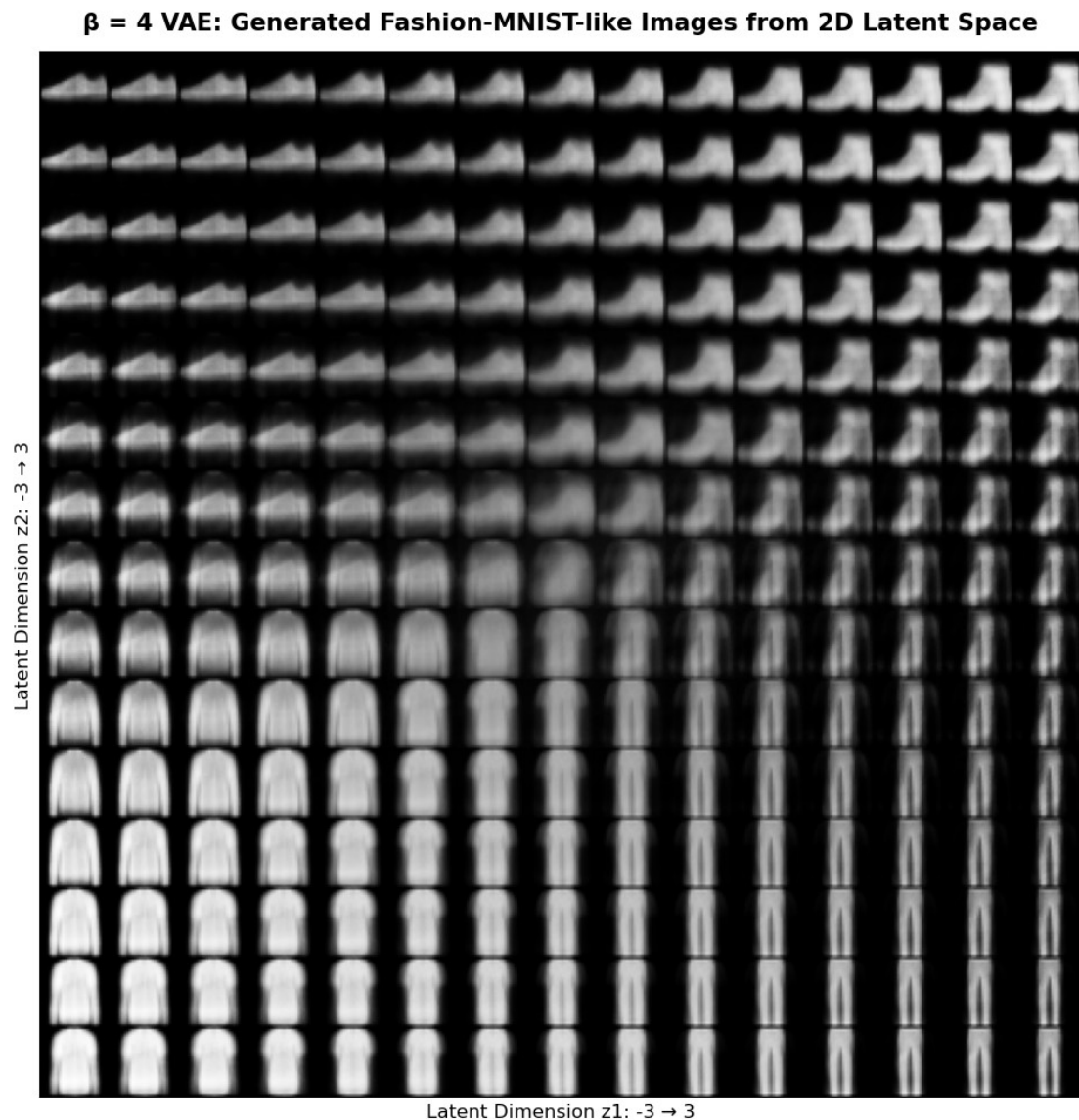


Figure 6: VAE generated images from sampled 2D latent coordinates.

9. Sparse Autoencoder

A Sparse Autoencoder forces only a few neurons to be active at a time. This encourages cleaner and more interpretable features. Tutor material lists sparse autoencoders as useful for feature extraction and interpretable representations.

Simple example: instead of using many pens to draw a picture, the model must use only a few important pens. This forces it to learn the most important patterns.

10. Denoising Autoencoder

A Denoising Autoencoder learns to reconstruct clean data from noisy input. The input is corrupted with noise, but the target remains the clean original. This is useful for image cleaning, robust features, and noisy sensor or ECG signals.

Simple flow: noisy image -> encoder/decoder -> clean reconstructed image.

11. Recurrent Autoencoder

A Recurrent Autoencoder is used for sequence data where order matters, such as ECG signals, time series, text, or sensor readings. It can use RNN or LSTM layers to compress and reconstruct sequences.

Simple example: a heartbeat signal is a sequence. If the order changes, the pattern changes. A recurrent autoencoder learns the normal sequence structure.

12. Backpropagation and Gradient Descent

Backpropagation finds how each trainable weight contributed to the loss. Gradient descent updates those weights to reduce the loss. Adam is a practical optimiser that performs these updates efficiently.

Concept	Simple Meaning
Loss	How wrong the model output is.
Backpropagation	Calculates how each weight should change.
Gradient descent	Updates weights step by step to reduce loss.
Optimizer	Algorithm that applies weight updates.

13. Applications of Autoencoders

Application	Simple Explanation	Example
Dimensionality reduction	Compress large data into fewer values.	784 pixels -> 32 bottleneck values.
Anomaly detection	Abnormal data reconstructs badly and gives high error.	Fraud, machine faults, unusual traffic.
Feature extraction	Use encoder outputs as learned features.	t-SNE/UMAP visualisation.
Image processing	Reconstruct, denoise, or compress images.	Fashion-MNIST reconstruction.
Recommendation systems	Learn compressed user/item patterns.	Predict user preferences.
Data augmentation	Generate synthetic data using VAEs.	VAE-generated fashion-like images.

14. ECG and Anomaly Detection Context

For ECG anomaly detection, an autoencoder can be trained on normal heartbeat signals. Normal signals reconstruct well; abnormal signals reconstruct badly and produce higher reconstruction error. The same idea applies to inventory anomalies, unusual sales patterns, website traffic spikes, and sensor faults.

15. Advantages and Limitations

Advantages	Limitations
Can learn without labels.	Reconstructions can be blurry or imperfect.
Useful for compression and feature extraction.	Latent space can be hard to interpret.
Works across images, text, time series, and signals.	Powerful models can overfit.
Can support anomaly detection and generation.	Not always better than task-specific supervised models.

16. Glossary of Important Terms

Term	Simple Meaning	Where It Appears
Autoencoder	Neural network that reconstructs its own input.	All notebooks
Encoder	Part of model that compresses input.	Dense AE, Conv AE, VAE
Decoder	Part of model that rebuilds input from compressed code.	All notebooks
Bottleneck	Small compressed representation inside an autoencoder.	Dense AE, Conv AE
Latent space	Hidden compressed space where data is represented.	VAE, feature plots
Reconstruction	Model output that tries to copy the original input.	All notebooks
Reconstruction loss	Error between original and reconstructed input.	All notebooks
MSE	Mean Squared Error; average squared mistake.	Evaluation
MAE	Mean Absolute Error; average direct mistake.	Theory
Binary Cross-Entropy	Loss often used for 0-1 pixel reconstruction.	Dense AE, Conv AE, VAE
Dense layer	Fully connected layer where each neuron connects to previous layer.	Dense AE, VAE
Conv2D	Layer that scans images with filters to learn local patterns.	Conv AE
Filter/kernel	Small scanning window used by Conv2D.	Conv AE
MaxPooling	Reduces spatial size while keeping strong signals.	Conv AE
UpSampling	Expands feature maps during decoding.	Conv AE
Feature extraction	Using learned compressed features for analysis.	Intermediate notebook
t-SNE	Visualises high-dimensional features in 2D.	Intermediate notebook
UMAP	Another 2D visualisation method for high-dimensional features.	Intermediate notebook
VAE	Variational Autoencoder; autoencoder with probability-based latent space.	Expert notebook
z_mean	Centre of latent distribution.	VAE
z_log_var	Log variance; spread of latent distribution.	VAE
ϵ	Random noise used in VAE sampling.	VAE
z	Sampled latent point passed to decoder.	VAE
Reparameterisation trick	Sampling method $z = \mu + \sigma \times \epsilon$.	VAE
KL divergence	Loss that regularises latent space.	VAE
β	Weight controlling strength of KL loss.	beta-VAE
beta-VAE	VAE where β changes reconstruction vs regularisation balance.	Expert notebook
Overfitting	Model memorises training data and generalises poorly.	Training interpretation
Validation loss	Loss on unseen validation data.	Training plots
Gradient	Direction showing how weights should change.	Backpropagation
Backpropagation	Process of calculating gradients through the network.	Training
Optimizer	Algorithm that updates weights.	Training
Adam	Common optimiser used for neural networks.	Training
Sparse Autoencoder	Autoencoder that encourages few active	Tutor theory

Term	Simple Meaning	Where It Appears
	neurons.	
Denoising Autoencoder	Autoencoder trained to reconstruct clean data from noisy input.	Tutor theory
Recurrent Autoencoder	Autoencoder for sequence/time-series data.	Tutor theory
Anomaly detection	Finding unusual data using high reconstruction error.	Applications
ECG	Heartbeat electrical signal over time.	Optional tutor context

17. References and Source Notes

- LD6_Beginner_Dense_Autoencoder_MNIST.ipynb - completed beginner notebook and generated outputs.
- LD6_Intermediate_Conv_Autoencoder_FashionMNIST.ipynb - completed intermediate notebook and generated outputs.
- LD6_Expert_Variational_Autoencoder_FashionMNIST.ipynb - completed expert VAE notebook and generated outputs.
- MVD/educx Module 2 Machine Learning exercise collection - Learning Day 6 Autoencoders Fundamentals task structure.
- Tutor Autoencoder PDF/PPT - autoencoder definition, components, variants, losses, applications, advantages, limitations, and optional ECG task.